

ROSEAU, MN, USA

WMO: 727477

Lat: 48.856N

Lon: 95.697W

Elev: 323

StdP: 97.51

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 04970

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -31.3	(c) -28.0	(d) -35.0	(e) 0.1	(f) -30.6	(g) -32.1	(h) 0.2	(i) -28.3	(j) 11.3	(k) -9.6	(l) 10.2	(m) -10.3	(n) 2.1	(o) 270	(p) 0.638

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.7	30.9	23.6	28.7	22.0	27.4	20.9	24.6	28.8	23.3	27.5	22.0	25.9	5.2	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.9	18.4	27.4	22.2	17.5	26.6	20.9	16.2	25.4	76.3	28.9	71.1	27.1	65.9	25.5	29.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 10.7	(b) 9.1	(c) 8.2	(e) -34.4	(f) 33.1	(g) 3.4	(h) 1.9	(i) -36.8	(j) 34.5	(k) -38.8	(l) 35.6	(m) -40.7	(n) 36.6	(o) -43.1	(p) 38.0		
(a) 10.7	(b) 9.1	(c) 8.2	(e) -34.9	(f) 26.7	(g) 3.6	(h) 1.9	(i) -37.5	(j) 28.1	(k) -39.7	(l) 29.2	(m) -41.7	(n) 30.3	(o) -44.3	(p) 31.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.3	-14.3	-12.1	-4.3	4.6	11.7	17.4	20.0	18.8	14.5	6.8	-2.0	-10.6		
	DBStd	13.44	8.55	7.62	7.63	5.81	4.83	3.74	3.13	3.32	4.86	4.92	6.40	7.64		
	HDD10.0	3172	754	619	444	178	38	1	0	0	14	123	360	640		
	HDD18.3	5304	1013	853	701	413	212	61	19	34	134	357	609	898		
	CDD10.0	1085	0	0	1	15	89	222	309	273	148	26	1	0		
	CDD18.3	175	0	0	0	0	6	32	70	49	18	1	0	0		
	CDH23.3	1808	0	0	0	7	90	316	675	508	201	12	0	0		
	CDH26.7	465	0	0	0	1	17	75	182	134	54	2	0	0		
(14) Wind	WSAvg	3.7	3.6	3.6	3.8	4.2	4.2	3.7	3.2	3.2	3.7	4.0	4.0	3.6		
(15) Precipitation	PrecAvg	532	17	12	15	36	63	103	84	81	63	38	17	19		
	PrecMax	761	78	32	43	78	215	235	184	279	176	112	69	45		
	PrecMin	366	0	0	0	0	6	15	27	18	8	2	0	0		
	PrecStd	86	17	8	12	24	42	49	38	56	42	27	15	12		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.9	6.0	17.1	23.8	28.7	31.4	32.6	32.5	31.2	24.9	16.1	4.8		
		MCWB	3.2	3.9	12.0	14.3	19.1	23.9	25.1	24.6	23.0	17.4	11.1	2.5		
	2%	DB	2.2	2.9	11.2	20.3	26.1	28.7	30.2	29.8	27.9	21.0	12.0	2.6		
		MCWB	1.2	1.9	7.9	12.0	16.2	21.8	23.9	22.4	20.4	15.2	8.9	1.4		
	5%	DB	0.1	1.4	7.5	17.3	23.0	27.1	28.1	27.7	26.0	17.5	8.9	1.3		
		MCWB	-0.7	0.6	5.2	10.4	14.9	20.2	22.3	21.3	18.9	12.8	6.6	0.3		
	10%	DB	-2.4	-0.9	5.0	13.9	21.1	24.9	27.3	26.3	22.8	14.1	6.4	-0.8		
		MCWB	-3.2	-1.3	3.2	8.7	14.3	18.8	21.7	20.3	17.4	10.6	4.2	-1.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.1	4.0	12.3	15.1	21.1	25.8	26.8	26.1	24.0	18.6	11.8	3.0		
		MCDB	4.6	5.1	15.3	21.5	26.1	29.3	30.8	31.2	29.5	22.1	14.8	4.2		
	2%	WB	0.7	2.4	7.9	12.7	18.5	23.2	25.0	24.0	21.8	15.8	8.8	1.6		
		MCDB	1.9	3.3	10.3	18.6	23.7	27.4	28.8	28.2	26.2	20.2	11.4	2.6		
	5%	WB	-0.8	0.7	5.2	11.0	16.5	21.4	23.7	22.5	19.9	13.1	6.8	0.3		
		MCDB	-0.1	1.6	7.3	16.4	20.9	25.1	27.5	26.8	24.3	16.4	8.8	1.3		
	10%	WB	-3.1	-1.2	3.4	9.2	14.9	19.7	22.3	21.2	18.3	11.2	4.4	-1.2		
		MCDB	-2.4	-0.7	5.0	14.1	19.4	23.8	26.1	25.2	22.1	14.4	6.7	-0.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.6	10.3	10.1	11.0	12.0	10.9	11.7	12.5	12.3	9.7	8.0	8.4		
		MCDBR	9.3	9.0	10.8	16.2	15.8	13.7	13.3	14.4	15.5	14.4	11.5	8.6		
	5% WB	MCWBR	8.5	8.0	8.3	9.7	8.4	8.0	8.4	8.7	9.1	9.1	8.4	7.8		
		MCWBR	9.1	8.6	10.6	14.6	13.0	11.7	12.2	12.9	13.5	13.2	10.7	8.4		
(40) Clear-Sky Solar Irradiance	taub	taub	0.265	0.300	0.294	0.344	0.361	0.377	0.381	0.375	0.347	0.303	0.276	0.259		
		taud	2.284	2.180	2.404	2.374	2.393	2.374	2.380	2.401	2.471	2.560	2.495	2.365		
	Ebn at Noon	Ebn at Noon	800	841	927	905	899	884	874	863	857	850	798	765		
		Edh at Noon	71	101	96	111	114	117	114	107	90	70	59	58		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.40	2.61	3.81	4.81	5.40	5.70	6.09	5.07	3.74	2.21	1.25	1.00		
	RadStd	RadStd	0.15	0.20	0.37	0.43	0.35	0.45	0.28	0.36	0.29	0.32	0.15	0.11		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (27 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page